Reaction of zinc with iodine

This experiment involves the **synthesis** of a **metal salt** by direct reaction of a **metal** and a **non-metal. Zinc** powder is added to a solution of **iodine** in ethanol. An **exothermic** reaction occurs, forming **zinc iodide**, which can be obtained by evaporating the solvent.

$$Zn + I_2 \rightarrow ZnI_2$$

This experiment illustrates the difference between metallic and non-metallic elements and their reaction to form a compound with new properties.

Materials
Apparatus and chemicals
Eye protection
Each group (or demonstration) requires:
3 x Test-tubes and bungs
Test-tube rack
Measuring cylinder (10 mL)
Electric kettle
Small filter funnel
Filter paper
Pasteur pipette
Thermometer (0–100 ^{o}C)
Spatula
Watchglass
Weighing boat or suitable container for zinc powder

Chemicals
Solid iodine (Harmful, Dangerous for the Environment), about 0.5 g
Zinc powder (Highly Flammable. Contact with water can also release a flammable gas, dangerous for the environment), about 0.5 g
Ethanol (Highly Flammable) about 5 mL

Procedure
SAFETY: Wear eye protection
a Measure out 5 mL of ethanol using a measuring cylinder. Place a thermometer in the ethanol and record the temperature.
b Add the ethanol to 0.5 g of powdered iodine in a test-tube. Stir carefully, using the thermometer, to dissolve the iodine. The solution should be dark brown. Note the temperature.
c When all the iodine has dissolved, slowly add the zinc powder using a spatula and stir the mixture with the thermometer. The temperature should rise, indicating an exothermic reaction.
d When the reaction is finished, the color of the iodine should have faded and excess zinc will be left. If not, add further small amounts of zinc powder and stir until the brown color due to iodine has gone.
e Filter the solution into another test-tube.
f Using a pipette, transfer a few drops of the filtrate on to a watch glass and allow the solvent to evaporate. This can be sped up by placing the watch glass on a beaker containing some hot water. Zinc iodide will be left as a white solid.

What's going on?

In this reaction a metal and non metal are being reacted together to form a new substance with different properties. The new substance that is formed is called an ionic substance. This is because in this reaction the two elements are going through a chemical change where they are being transformed from neutral atoms to form positive and negative ions.

The zinc metal is losing two electrons to form Zn^{2+} and the iodine molecules are gaining electrons to form two I^- ions. As you will see the physical appearance of the two reactants will change with the formation of the new products, along with their chemical properties.

Observations

- Note the physical appearance of each substance

- Note what occurs during the reaction

- Note the appearance of the new substance

Questions

1. How do you know a reaction has occurred?
2. What do you think exothermic means?
3. How have the physical and chemical properties of each reactant changed?
4. What is an ion?

* This experiment is an assessment item and is to be handed in as a written up lab report, containing an aim, method, procedure, observations, discussion and conclusion. It is due next Friday 16/5/2008.